

RISC-V 5-Stage Pipelined Processor Documentation

Supported Instructions

Category	Instruction	Description
R-Type	ADD,SUB,OR,AND	Operates on contents of two source registers
I-Type	ADDI	Adds register content with immediate generated value
Load	LW	Loads a 32-bit word from data memory
Store	SW	Stores a 32-bit word to data memory
Branch	BEQ	Branches when two register values are equal

Instruction Execution Summary

PC	Instruction	Operation	Expected Result
0x00	add x3,x1,x2	$x3 = x1 + x2$	$x3 = 30$
0x04	sub x4,x3,x1	$x4 = x3 - x1$	$x4 = 20$
0x08	add x5,x4,x3	$x5 = x4 + x3$	$x5 = 50$
0x0C	lw x6,0(x1)	Load word	$x6 = 50$
0x10	add x7,x6,x5	$x7 = x6 + x5$	$x7 = 100$
0x14	sw x7,4(x1)	Store word	Data_mem[14] = 100
0x18	lw x8,4(x1)	Load stored word	$x8 = 100$
0x1C	add x0,x1,x2	Attempt write to x0	x0 remains zero
0x20	add x9,x0,x1	$x9 = x1$	$x9 = 10$
0x24	beq x1,x2,+8	Compare registers	Branch not taken
0x28	beq x1,x1,+8	Compare registers	Branch taken
0x2C	add x10,x10,x10	Instruction after branch	Skipped after flush
0x30	add x11,x7,x8	$x11 = x7 + x8$	$x11 = 200$

Hazard Handling

Hazard Type	Instruction Sequence	Expected Hardware Behaviour	Verified By	Status
RAW	add → sub	Forwarding Unit resolves dependency	Waveform	Passed
RAW	sub → add	Forwarding Unit resolves dependency	Waveform	Passed

Load-Use	lw → add	Hazard Detection inserts stall, forwarding after load	Waveform	Passed
Store Dependency	add → sw	Latest ALU value stored	Waveform	Passed
Control Hazard	beq not taken	Sequential execution	Waveform	Passed
Control Hazard	beq taken	Pipeline flush and PC redirected	Waveform	Passed
Zero Register	write x0	Write ignored	Waveform	Passed

Verification Summary

Feature	Verification Method	Status
ALU Operations	Waveform	Passed
Register File	Waveform	Passed
Instruction Fetch	Waveform	Passed
Instruction Decode	Waveform	Passed
Load/Store	Waveform	Passed
Forwarding Unit	Waveform	Passed
Hazard Detection Unit	Waveform	Passed
Branch Flush	Waveform	Passed
Pipeline Registers	Waveform	Passed
SystemVerilog Assertions (SVA)	Not yet implemented	Planned

Forward selection

At t=400 Instruction 1 is fetched

At t=600 Instruction 1 is decoded and Instruction 2 is fetched.

At t=800 Instruction 1 is in execution and Instruction 2 is decoded and Instruction 3 is fetched. Here destination register (rd) of Instruction 1 is equal to source register 1(rs1) hence rs1_sel is 01 which means the execution result which supposed to be stored in rd is supposed to be used in instruction 2 rs1.

At t=1000 Instruction 1 is in memory section Instruction 2 is in execution and Instruction 3 is decoded rs2_sel is 10 and rs1_sel is 01 rs2_Sel 10 meaning memory stage rd of instruction 1 is equal to be used in rs2 of instruction 3.

Refer images Forward_sel_Wav and Forward sel_Transcript.

Hazard Detection

At t=1000 instruction 4 is fetched

At t=1200 instruction 5 is fetched

At t=1400 stall signal is generated because source register of instruction 5 needs updated value of rd in Instruction 4 which needs memorywrite operation hence a after stall is generated the pc incrementer is stopped for 1 cycle. Stall signal brought to 0.

At t=1600 the value obtained is passed to rs1 by rs1_sel signal which 10 and pc resumes its normal operation

Refer images Hazard_Detect_Wav and Hazard_Detect__Transcript.

Branch Operation

At t=2800 instruction 11 is executed and since result is zero zero_flag is raisen and branch is also high the pc_branch = pc+immediate value passed here then the 2 instruction passed are redundant before execution of instruction 11 and psel is made high.

At t= 3000 here it is dealt by making control signals of instruction 12 to be 0 while coming to execution and during fetch operation of instruction 13 it is made all 32 bits 0 a special NOP (Non operation Opcode) is introduced for inducing all control signal 0 .

At t=3200 Hence it can be noticed again a zero_flag is raisen for instruction 13 at execution which is manipulated as all 32 bits zero.pc updates .but since pc=pc+8 it again fetches the instruction 13 at fetch stage.

Refer images branch_Wav and branch__Transcript.

Zero Register

At t=2000 instruction 8 is decoded where it sees rd as x0 hence the RegWrite is made 0 in execution part.

Refer image Overall Waveform in transcript x0 is not updated

Here all internal signals are not made to be appeared in overall waveform and specific important detection logic are made with essential internal signals in waveform . Psel is for branch operation and stall is for hazard detection.